

# Telemedicine Usage Trends in the United States of America between 2020-2024

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## Introduction

Telemedicine or telehealth, is the remote delivery of healthcare services using technology. With the impact of COVID-19, telemedicine adoption increased due to lockdowns and need for physical distancing [5]. An increase in telehealth usage during the pandemic has raised important questions regarding the effectiveness, accessibility, and long-term viability of the service. In this project, we explore several key aspects of telehealth trends in the United States between 2020 and 2024 using Medicare beneficiary data. Our research addresses the following questions:

1. How has telehealth use changed over time? Specifically, is there a significant difference in the average percentage of telehealth users between the years 2020–2024?
2. Is there a significant difference in the average number of telehealth visits between rural and urban areas? How does usage vary across states?
3. Is there a significant difference in the frequency of telehealth use across different age groups? Are younger beneficiaries more likely to use telehealth than older ones?

Understanding these patterns is crucial for policymakers, healthcare providers, and public health researchers. By identifying disparities and usage trends, we can better inform healthcare strategies to improve accessibility as well as adapt to future trends in technology usage.

## Related work

Prior research on telemedicine usage during the COVID-19 pandemic has shown that telemedicine helped ensure continuity of care while reducing virus exposure. This was especially beneficial for individuals managing chronic illnesses and mental health conditions [5].

However, several studies have also highlighted issues of inequality and accessibility within telehealth systems. For example, research published in *The Journal of Rural Health* found that geographic disparities contributed to unequal access to internet and digital technology, which limited telehealth adoption in rural communities [6]. Similarly, Lam et al. (2022) identified age-related differences in telehealth usage, noting that older adults were less likely to use video visits due to lower digital literacy and technological barriers—raising concerns about age-based accessibility gaps [3].

Given these findings, a deeper understanding of how telehealth usage varies over time, by age, and by geographic location can help public health researchers and policymakers develop strategies to improve telemedicine accessibility for all populations in the United States.

## Data

The first dataset we used is the “[Medicare Telehealth Trends](#)” dataset, collected by the Centers for Medicare & Medicaid Services (CMS) of the United States government. This dataset provides comprehensive information on Medicare beneficiaries who used telehealth services between January 1, 2020, and September 30, 2024. The data is segmented by year, quarter, geographic location (national, state, urban, rural), and various demographic factors such as age, race, and sex [2].

The second dataset is titled “[Telehealth Use in the Last 4 Weeks](#),” made available by the U.S. Census Bureau through Data.gov. This dataset is based on household surveys and provides information on the percentage of adults who used telemedicine services in the past four weeks. It offers insights into telehealth usage patterns across time and various demographic groups [4].

We selected these datasets based on their relevance, coverage, and credibility. Both directly support our research objectives, which include analyzing telehealth usage by time period, geographic location, and age. The CMS dataset is particularly valuable for its detailed, longitudinal structure, which supports temporal trend analysis over multiple years. Meanwhile, the Census Bureau dataset offers a broader view beyond Medicare recipients, allowing us to consider telehealth trends within the general U.S. adult population. Most importantly, both datasets are sourced from reputable federal agencies, ensuring high reliability and data validity.

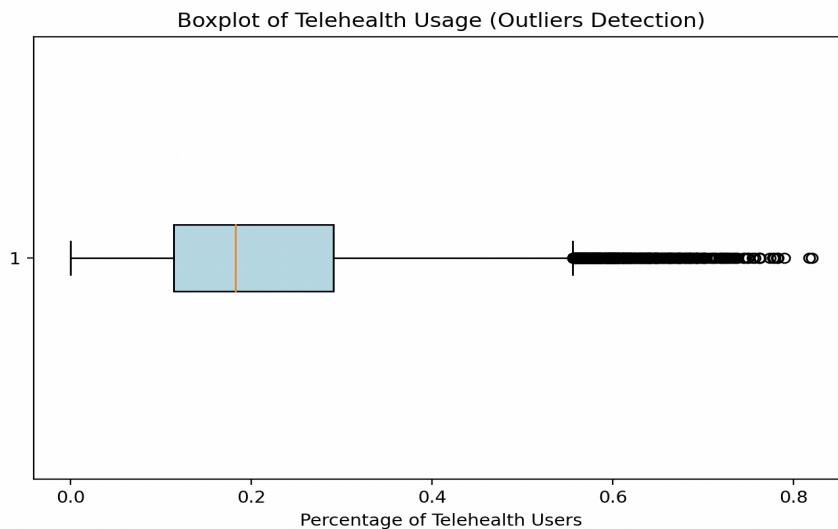
In order to prepare the data for analysis, we performed a series of data cleaning steps and exploratory data analysis (EDA) prior to addressing our research questions. First, we identified and removed columns that were not relevant to our analysis, referencing the CMS data dictionary for the *Medicare Telehealth Trends* dataset as well as feature definitions provided in the U.S. Census Bureau dataset website [1].

For the “Telehealth Use in the Last 4 Weeks” dataset, we dropped the following columns: 'Indicator', 'Phase', 'Confidence Interval', 'Quartile Range', and 'Suppression Flag'. For the *Medicare Telehealth Trends* dataset, we removed 'Bene\_Mdcr\_Entlmt\_Stus' and 'Total\_PartB\_Enrl'. These columns were excluded due to their irrelevance to our primary variables of interest.

After dropping irrelevant columns, we checked for missing values and applied the **dropna()** function to remove rows containing nulls, ensuring that only complete and reliable data remained. In the case of the “Telehealth Use in the Last 4 Weeks” dataset, we then organized the data by splitting it into multiple smaller datasets based on the "Group" column. These groups represent different demographic or categorical breakdowns of telehealth usage, such as by age, state, or national estimate. To align with our research focus on usage by service type, geographical location, and age, we retained the following three groups: National Estimate, By

Age, and By State. Groups such as By Gender and By Education were excluded, as they did not directly relate to our research questions.

Following data cleaning, we performed EDA on the cleaned *Medicare Telehealth Trends* dataset (saved as `medicare_cleaned.csv`) to better understand the structure and variability of the data. The dataset contains 23,626 rows and no missing values. One of our main variables of interest is `Pct_Telehealth`, which indicates the percentage of telehealth users. To analyze its distribution, we created a box plot.



The box plot allowed us to visualize the central tendency, spread, and presence of outliers within the data. Given that the underlying distribution of this variable is unknown, the box plot served as a useful tool for summarizing the data visually.

From the plot, we observed a significant number of outliers above the upper whisker, indicating unusually high telehealth usage in certain records. The distribution is right-skewed, with a median around 0.18. While outliers are often removed in statistical analysis to improve accuracy, in this case, they may reveal important insights. Rather than excluding them, we interpret these values as potentially meaningful, as they could reflect higher-than-average telehealth adoption influenced by factors such as the COVID-19 pandemic, state-level healthcare policies, demographic characteristics, or differences in broadband access. These outliers may be key to answering our research questions regarding variability across time, geography, and population groups.

## Methodology

For evaluating telemedicine usage trends by geographical location, we analyzed both datasets separately and combined statistical testing, descriptive analysis, and visualization to address our research question. First, we analyzed **medicare\_cleaned.csv** by filtering the **Bene\_RUCA\_Desc** column to separate urban and rural groups. We then calculated the mean number of telehealth visits for each group. To assess whether the differences were statistically significant, we performed an independent two-sample t-test and compared the resulting p-value to our significance level of 0.05. We chose to use a t-test because we were comparing means between two independent groups, and it is appropriate given the continuous nature of the **Total\_Bene\_Telehealth** variable. However, there are some limitations to this approach as it assumes normality for the t-test and does not control for confounding variables like age, income, or health status.

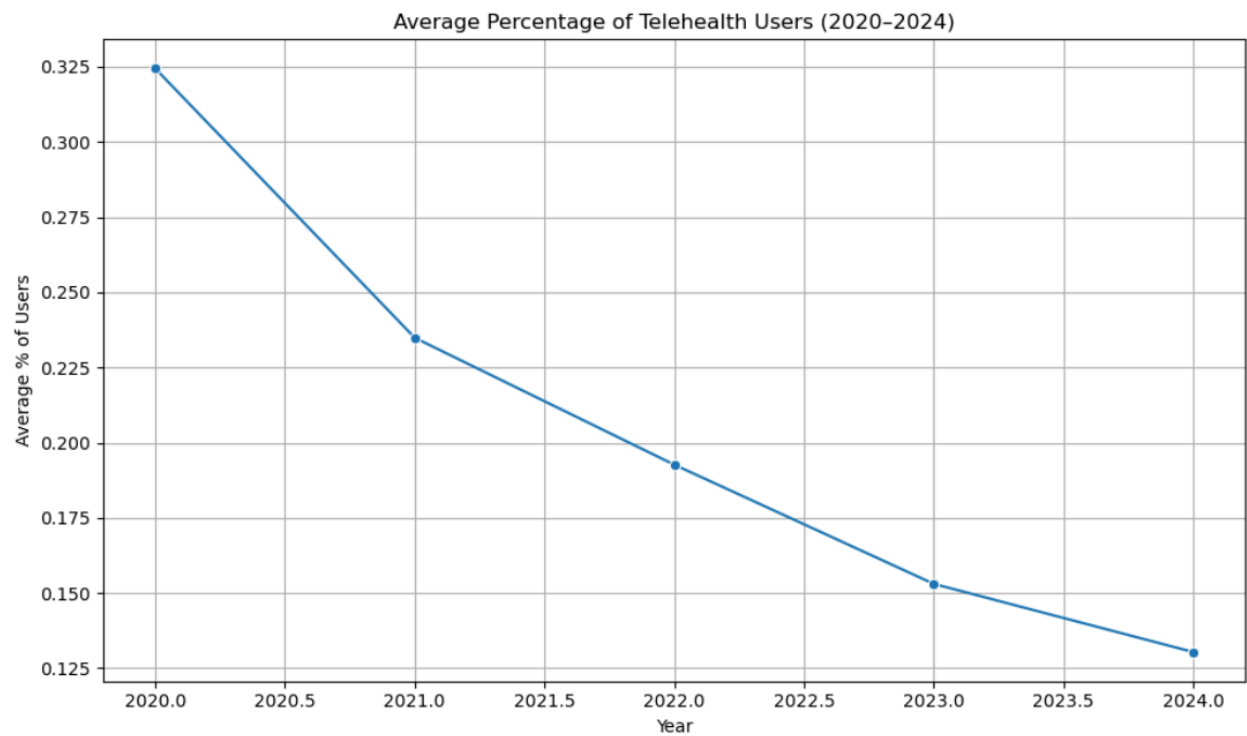
Next, we examined state-level telemedicine adoption using the **telemedicine\_group\_By\_State.csv** dataset from the U.S. Census Bureau. We calculated the average percentage of adults using telehealth by state, identified the highest and lowest usage states, and visualized the results using a bar chart. Aggregating and ranking usage by state allowed us to explore broader geographic trends in telehealth adoption across the country.

To answer our third research question : “Is there a significant difference in the frequency of telehealth use across different age groups?”, we first analyzed **medicare\_cleaned.csv**. We filtered the dataset to retain only entries with a specific age group. The variable **Pct\_Telehealth**, representing the percentage of eligible beneficiaries in each age group who used telehealth, was our primary measure of usage. We grouped the data by **Bene\_Age\_Desc** and calculated the average telehealth usage percentage for each group. For **telemedicine\_group\_By\_Age.csv**, we grouped the data according to the age categories defined in the Subgroups column. The **Value** column represents the percentage of people in each group who use telehealth, which was our primary measure of usage. Finally, we calculated the average percentage of telehealth use for each age subgroup. In order to test whether the differences were statistically significant, we used a one-way ANOVA. ANOVA is appropriate for comparing the means of more than two groups, and is very robust for large sample sizes such as ours. Using this approach, we were able to evaluate patterns of telehealth use across age groups in the Medicare population.

Results and Discussion

**Research question 1:** How has telehealth use changed over time? Specifically, is there a significant difference in the average percentage of telehealth users between the years 2020–2024?

Our results show a clear decline in the average percentage of Medicare beneficiaries using telehealth services from 2020 to 2024. The usage dropped from approximately 32.4% in 2020 to just 13.0% in 2024. This suggests that telehealth usage surged during the COVID-19 pandemic but has steadily decreased in the following years.



Internal validity

The data used comes from official Medicare claims, which accurately records telehealth usage across various demographics. The measurement directly aligns with the research question and calculates usage as a percentage of eligible beneficiaries, making it a valid measure of what we intended to examine.

External validity

Since the data includes Medicare users across the entire U.S., it has strong generalizability for older adults and people with disabilities. However, it may not fully represent telehealth trends in the general population, especially younger individuals or those not covered by Medicare. Therefore, the findings are somewhat limited in generalizing to all healthcare users in the U.S.

**Research Question 2:** Is there a significant difference in the average number of telehealth visits between rural and urban areas? How does usage vary across states?

To investigate differences in telehealth usage between rural and urban areas, we analyzed data from **medicare\_cleaned.csv**. The average number of telehealth visits in rural areas was approximately 60,346, compared to 288,016 in urban areas. To assess whether this difference was statistically significant, we conducted an independent two-sample t-test. The resulting t-statistic was -8.43 with a p-value of  $8.42 \times 10^{-17}$ , which is well below our significance threshold of  $\alpha = 0.05$ . Therefore, we reject the null hypothesis and conclude that there is a statistically significant difference in average telehealth visits between rural and urban populations.

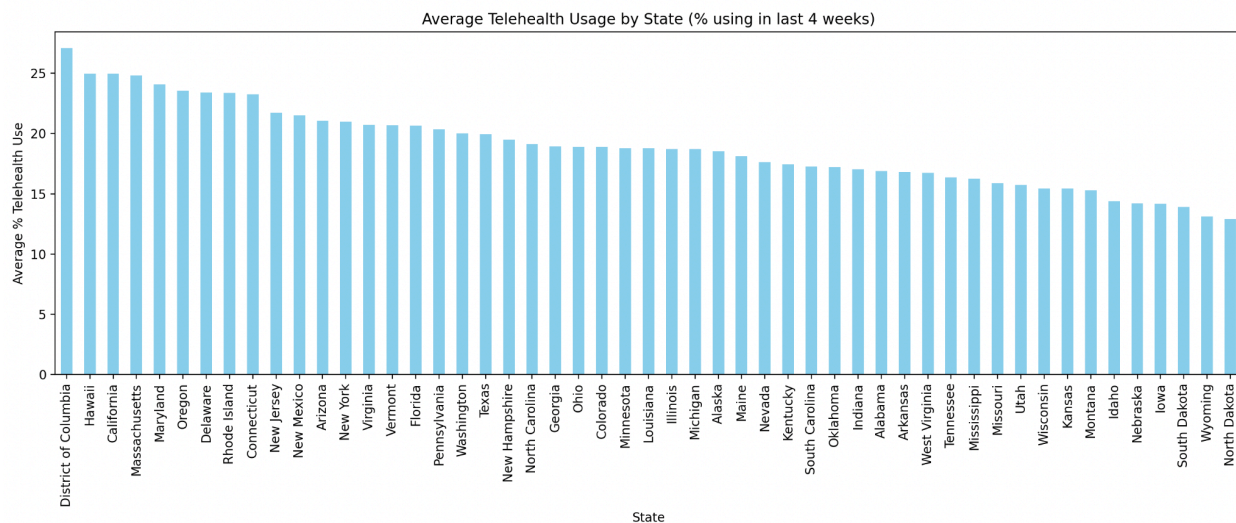
Next, to examine how telehealth usage varies across states, we analyzed the **telemedicine\_group\_By\_State.csv** dataset, which is adapted from the “Telehealth Use in the Last 4 Weeks” survey. We calculated the average percentage of adults using telemedicine in each state and identified the top and bottom five states by usage. The top five states were:

- **District of Columbia (27.10%)**
- **Hawaii (24.98%)**
- **California (24.96%)**
- **Massachusetts (24.82%)**
- **Maryland (24.09%)**

The five states with the lowest average usage were:

- **Nebraska (14.21%)**
- **Iowa (14.16%)**
- **South Dakota (13.90%)**
- **Wyoming (13.14%)**
- **North Dakota (12.90%)**

These results suggest that telehealth usage varies considerably across states, with generally higher adoption in urban or coastal regions and lower adoption in more rural or central areas.



## Internal Validity

Our measures are valid representations of the concepts in our research question, and the analysis closely aligns with the objective of assessing differences in telehealth usage. For our analysis of average telehealth usage, we used the **Total\_Bene\_Telehealth** column from the **medicare\_cleaned.csv** dataset, which directly measures the amount of telehealth visits from Medicare beneficiaries. The rural and urban classifications were based on the **Bene\_RUCA\_Desc** field, which reflects the rural/urban status defined by the mailing ZIP code and the Rural-Urban Commuting Area Crosswalk (RUCA), and aligns with how rural-urban classifications are determined in the U.S.

## External Validity

The Medicare dataset documents actual telemedicine usage from Medicare beneficiaries, which reflects real-world accuracy and is representative of usage within that specific population. However, since the Medicare dataset only covers Medicare beneficiaries—primarily older adults and people with disabilities—it may not be generalizable to the broader U.S. population. The Census-based telemedicine dataset includes the broader U.S. adult population, offering a wider perspective on telemedicine usage outside of Medicare and adding external validity to our analysis. While the Medicare dataset may not capture the full scope of telemedicine usage beyond the Medicare population, the telemedicine dataset complements this by documenting usage among the general population.

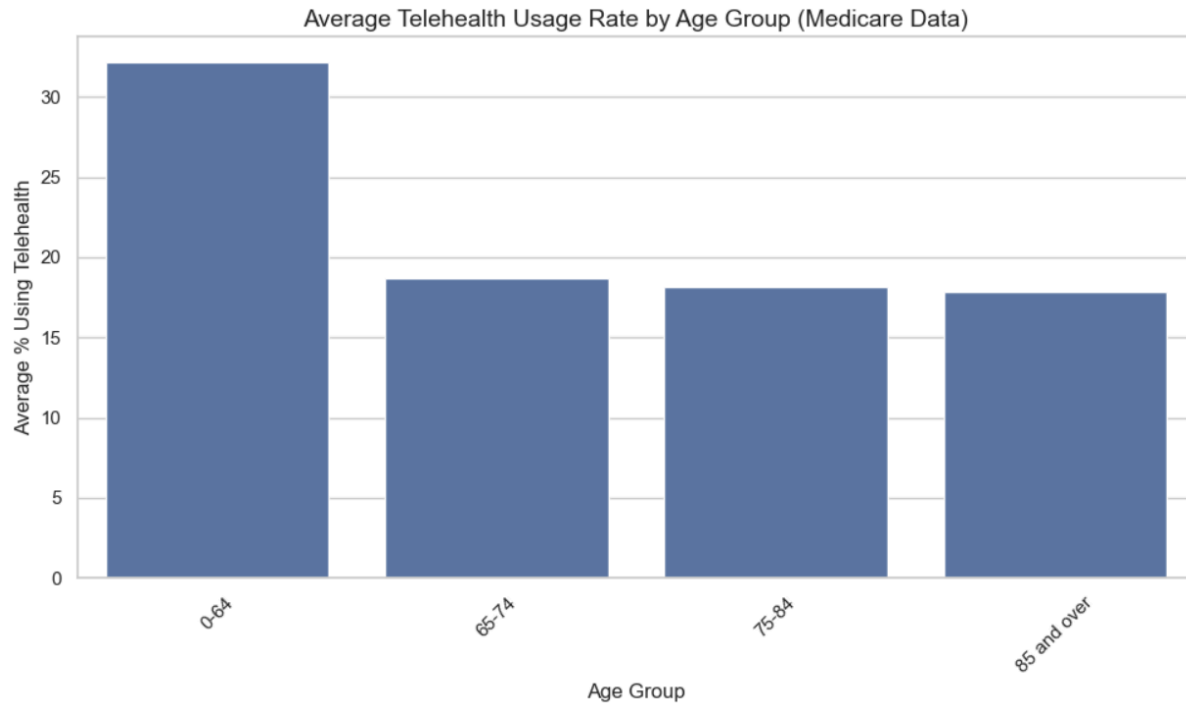
**Research question 3:** Is there a significant difference in the frequency of telehealth use across different age groups? Are younger beneficiaries more likely to use telehealth than older ones?

For **medicare\_cleaned.csv**, we have the results showing below:

- **0-64 (32.14%)**
- **65-74 (18.68%)**

- **75-84 (18.16%)**
- **85+ (17.84%)**

A bar chart showing the average telehealth usage rate by age group of this dataset is provided below:



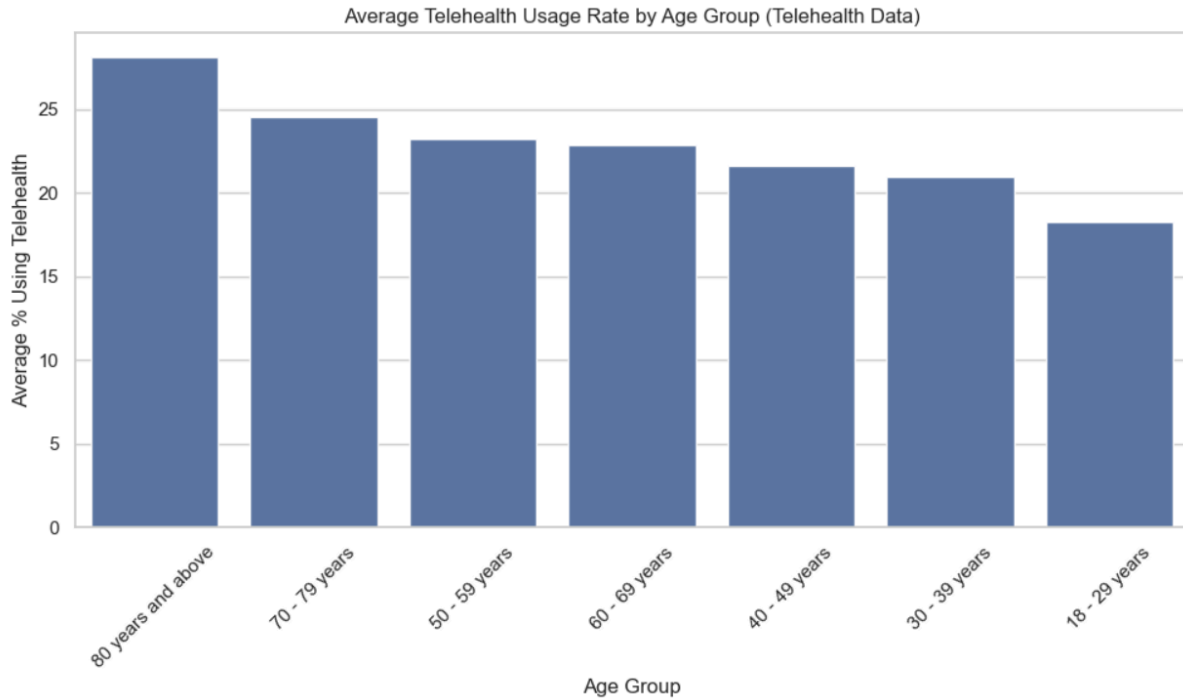
The p-value from one-way ANOVA is 1.395e-230, which is smaller than significance level of 0.05, suggesting that age group has a significant effect on the likelihood of telehealth usage within the Medicare population.

For **telemedicine\_group\_By\_Age.csv**, we have the results showing below:

- **18-29 (18.26%)**
- **30-39 (20.98%)**
- **40-49 (21.60%)**
- **50-59 (23.25%)**
- **60-69 (22.86%)**
- **70-79 (24.59%)**
- **80+ (28.15%)**

A bar chart showing the average telehealth usage rate by age group of this dataset is provided below:





The p-value from one-way ANOVA is  $3.840e-23$ , which is smaller than the significance level of 0.05, suggesting that telehealth usage differs significantly by age group.

#### Internal Validity:

The measure used **Pct\_Telehealth** is derived directly from Medicare claims data, which ensures a high degree of data accuracy and alignment with our research questions. And the **telemedicine\_group\_By\_Age.csv** directly measures the proportion of people in each age group who have used telehealth in the last four weeks. The dataset is large and structured to be broken down by age, making it well suited to our analysis.

#### External Validity:

Although the Medicare claims dataset provides the information based on the actual medical claims, its application is limited to the Medicare population, which is primarily elderly and disabled. Thus, the findings might not be fully generalizable to younger people or individuals with private insurance.

In comparison, the Census Bureau's dataset is derived from a nationwide representative survey and thus has higher external validity for a broader group of U.S. adults. However, since the data rely on self-reporting, individuals may have biases in recalling or interpreting their telemedicine usage.

## Discussion of results

The significant drop in telehealth usage after 2020 suggests that the surge was driven by the pandemic and may not be sustainable without ongoing policy support or user demand. This decline could impact healthcare accessibility, particularly for individuals who relied on telehealth during lockdowns or in rural areas. It raises concerns about whether telehealth will remain a viable and widely used option in the long term.

Based on our results from rural vs. urban telehealth usage, we found that urban areas have a significantly higher number of telehealth visits compared to rural areas. Our telehealth adoption analysis by state showed that usage varies substantially across states, with higher usage generally in urban or coastal areas and lower adoption in central areas. This is likely due to the fact that urban areas have better internet access, more healthcare infrastructure, and greater digital literacy. On the other hand, rural areas have fewer telehealth visits possibly due to limited internet access, less healthcare providers offering telehealth, and lower levels of digital literacy. This finding confirms that there is a statistically significant disparity in terms of telemedicine access based on geographical location, which answers our second research question. State-level insights provide more specific usage information and could be used to inform public health policy and improve healthcare technology infrastructure.

Younger Medicare beneficiaries (0-64) appear to be using telehealth services at a significantly higher rate than other age groups. This could be due to greater ease of technology, increased outpatient contact (mental health care) or cognitive restrictions to access. However, comparing this result to our second dataset, “Telehealth Utilization in the Last 4 Weeks” from the U.S. Census Bureau, we found an interesting contrast. This dataset, which is based on self-reports from a broader sample of U.S. adults, found that telehealth use was higher among older adults (ages 80+) and lower among the youngest adults (ages 18-29).

One limitation is that this dataset only includes Medicare beneficiaries, which may exclude key populations such as uninsured individuals or those with private insurance. Also, differences in how telehealth services are coded or reported across providers might introduce inconsistencies. It is also important to note that there is a temporal mismatch between the two datasets used. While the medicare dataset covers usage between 2020-2024, the Census-based telemedicine dataset only covers 4 weeks usage at various time points. This may introduce inconsistencies when comparing trends between the two. Data aggregation in both datasets may also mask variability within states, quarters, and subgroups. For example, urban vs. rural differences within a single state may not be captured when we aggregate by state. The Census-bureau telemedicine dataset also used self-reported data, which could be affected by factors such as response bias. Finally, external factors like policy changes, technology access, or regional healthcare availability are not fully captured in this analysis.

## **Future work**

Our findings show that the use of telemedicine in the United States differs significantly across time, location, and age groups. These results could help healthcare policymakers and providers focus their efforts more effectively. For instance, as telemedicine utilization declines after 2020, it may be helpful to address how policy and infrastructure can support more sustainable use. Again, with lower adoption rates in rural areas and in some states, our results could support future decisions to improve Internet access or expand telemedicine services in underserved communities.

Given more time, we wish to analyze differences in telemedicine use by race or ethnicity, income level, or type of insurance, as these factors could also affect telemedicine access and adoption. For example, even in urban areas, some groups may still face barriers due to economic or social factors that were not included in our current analysis.

We also noticed differences between the two datasets we used (Medicare claims and census data). Future projects could give more attention to in-depth comparisons of these data sources or use other data sources, such as electronic health records, broadband access maps, or mental health service availability, to provide a better explanation of utilization patterns. An additional possible extension is the use of time-series analyses to model how events such as policy changes or public healthcare crises affect the use of telehealth. This could help predict and prepare for future healthcare needs.

In conclusion, our study provides a foundation for understanding telehealth utilization, but further research is needed to explore more detailed factors and improve healthcare services for all populations.

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